

Adroit Technologies

Protocol Driver

Name	MyCity Email IMAP
DLL	MYCITY



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1. Overview

This driver supports IMAP for accessing and reading in MyCity RTU protocol packets and updating Real and Analog Agents.

1.1. Test Software

This driver has been tested using the following email services:

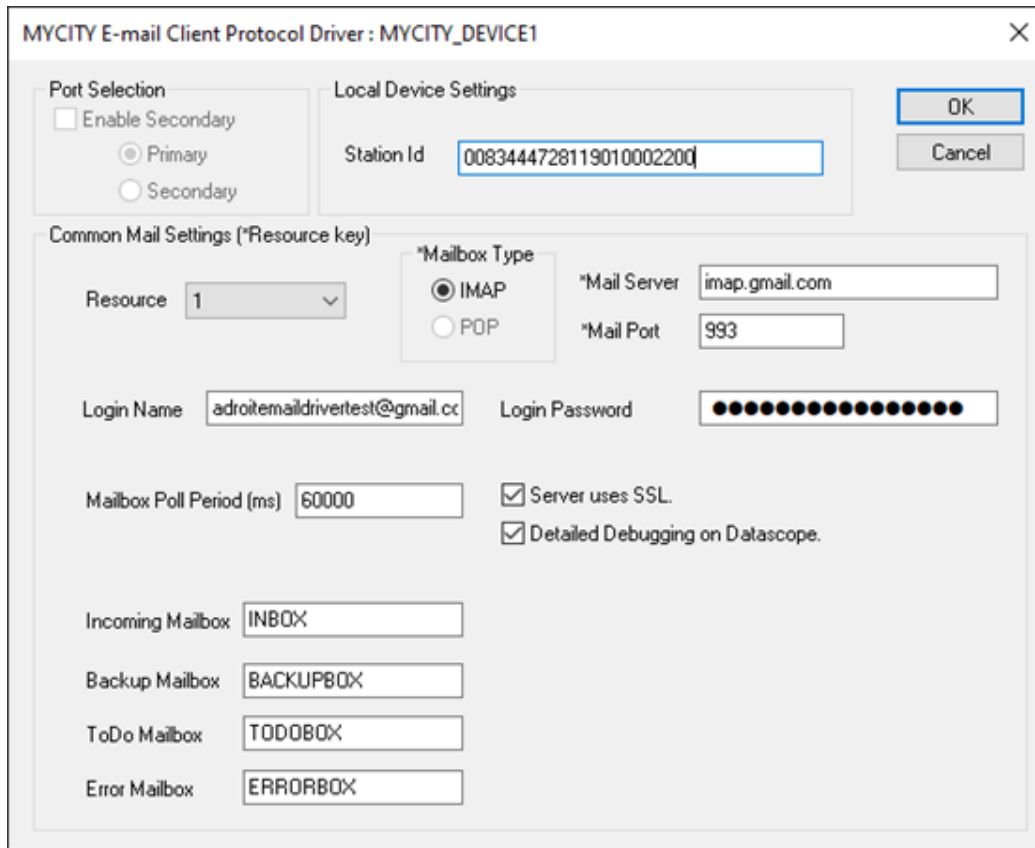
- Gmail
- Outlook

2. Wiring

Not Applicable.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. Common Mail Settings

*(The parameters identified by the * define a resource common to all devices that share those parameters. Any change to these parameters in this group will cause a change for all Adroit devices using this mailbox. Note: this will include other drivers such as the Adroit SMTP Email client driver.)*

Resource - An internal ID representing a mailbox configuration in Adroit that cannot be edited. When adding a new device, the last added resource will be used to pre-populate the mailbox settings (this is used to make the adding of multiple devices easier). To choose another resource, simply select it and click OK. To change anything the driver will ask if a new resource is to be created or the existing resource be changed for all devices using it.

***Mailbox Type** - IMAP is automatically selected (**POP is no longer supported**).

***Mail Server** - The name of the mail server.

***Mail Port** - 993 for IMAP (**143 is now no longer used**), 110 for POP (**no longer supported**).

Login Name - Username on the server.

Login Password - User password.

Mailbox Poll Period (ms) - Checking the mailbox for new mails will occur at this interval.

Server uses SSL - [On by default] Some POP and IMAP servers require SLL mode.

IMAP servers listen on port 143 typically, while **993** is used for SSL/TLS. **POP** servers listen on port 110 typically, while **995** is used for SSL/TLS (**no longer supported**.)

Detailed debugging to datascope - Each record updated to Adroit will be displayed.

Incoming Mailbox - Incoming account inbox name.

Backup Mailbox - Any emails that are successfully read from the inbox that are for a device that is started will be moved to this mailbox. No emails will ever be deleted from this box; therefore, it is up to the relevant user to perform manual deletion. (NB – this function is only allowed in IMAP mode).



PLEASE NOTE

Some servers do not allow multiple mailboxes on one account, but they usually allow subfolders, if this is the case: use dot notation.

For example:

INBOX.BACKUPBOX

ToDo Mailbox - When a valid email is received that is for a device that has not been started or is currently stopped will move to this mailbox. When a device is

Error Mailbox - When an invalid email is received and Adroit cannot process the incoming mail, the email will be moved to this mailbox. Emails here will not be checked again or deleted and must be done manually by the user. Any spam that has not been automatically filtered will also be moved here.

3.1.3. Local Device Settings

Station ID - This is the identifier code sent in the subject line of the email. You can use any fixed identifiable substring in the subject line.

For example: "Subject: F000001398 : 0082728922419110000100"

The StationID could be the last number including the colon

": 0082728922419110000100" or "F000001398" or "F000001398 : 0082728922419110000100" or the whole string.

4. Supported Addresses

A CSV row constitutes a record. The example email body consists of two records.

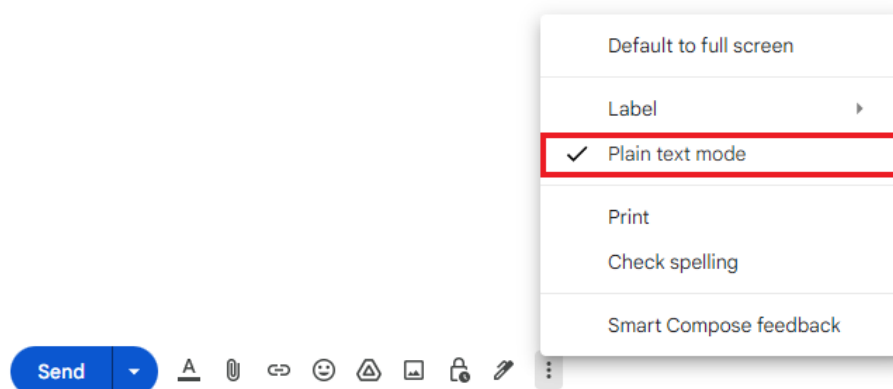
```
2022-05-01
01:15:00,3614.53979,24.00000,1.50000,131.00000,250.00000,0.00000,1.00000,
0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000,
0.00000,0.00000
2022-05-01
01:30:00,3629.57690,24.00000,0.00000,131.00000,250.00000,0.00000,1.00000,
0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000,
0.00000,0.00000
Checksum: 29208
```

The format follows the following structure:

Date[YY-MM-DD] Time[HH:MM:SS],Param1,Param2,...Param48

The maximum number of parameters/Param<x> fields is 48.

You can test this driver with the above example. The checksum field, although necessary to still have in the email body, does not have to be correct but make sure it still has an integer type, otherwise it will fail to validate and cause errors. If you use the above example and it is not working, make sure to only have a Carriage Return (CR) character followed by a Line Feed (LF) character between CSV rows. There should be no white character spaces (spaces, tab spaces, etc.) Then when copying to an email, ensure that you send the email using plaintext only. For example, with Gmail:

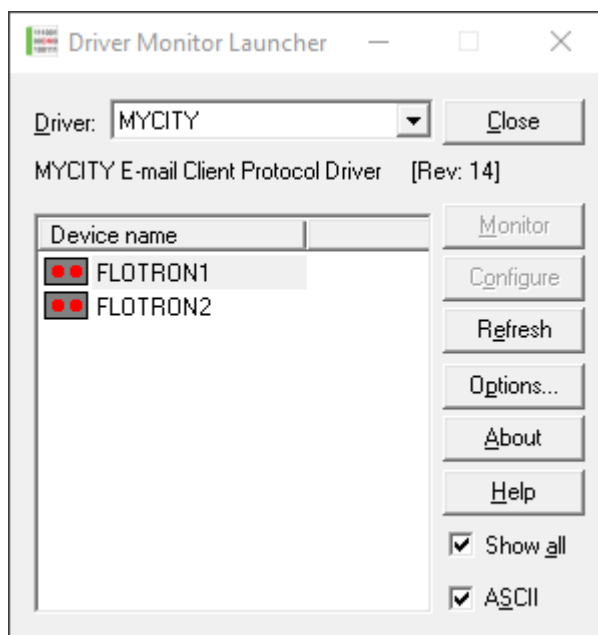


The address can only be used for Adroit Real Agents only. To scan an Adroit agent to a parameter, use the scan address Chan<x> as follows:

Email Data Parameter	Adroit Scan Address
Param1	Chan0
Param2	Chan1
...	...
Param48	Chan47

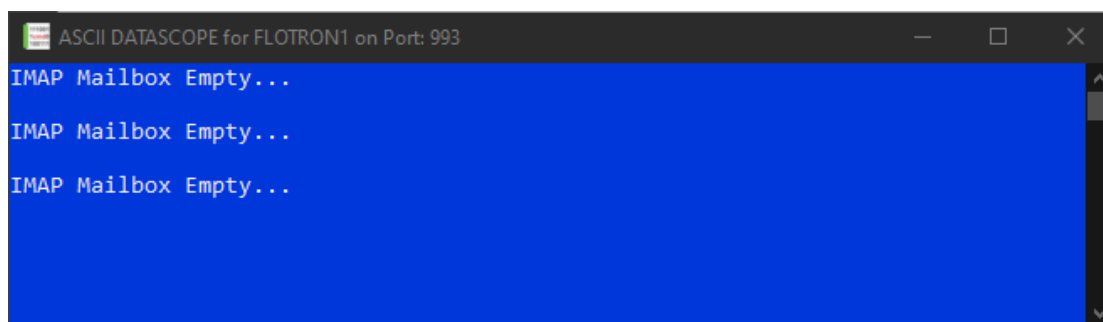
5. Protocol Definition and Considerations

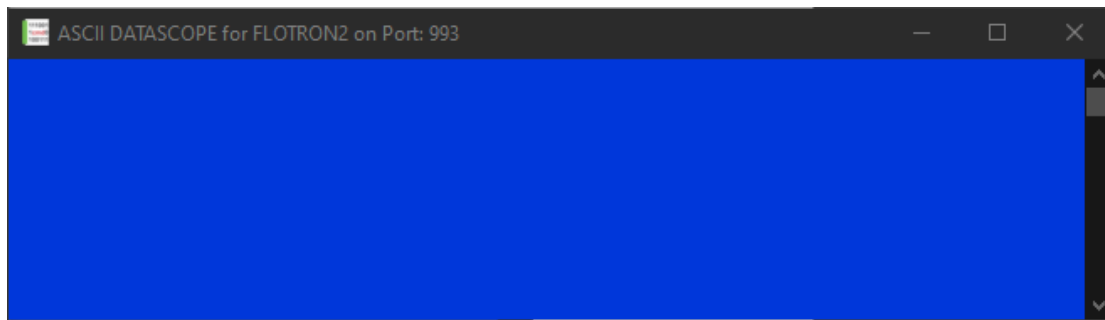
5.1. Datascope Considerations



When the Adroit Driver Monitor Launcher is examined, all devices will be seen. Note that one of them will have indicate more activity than the others. This is usually the top one in the list, which is the first one to attach in the agent server on a particular resource (mailbox). This is where all mailbox activity will be seen – call it the activity log. If a second mailbox resource is added, the first device to attach would be designated the “activity log” for that mailbox. It would not necessarily be at the top, so one would have to look for it.

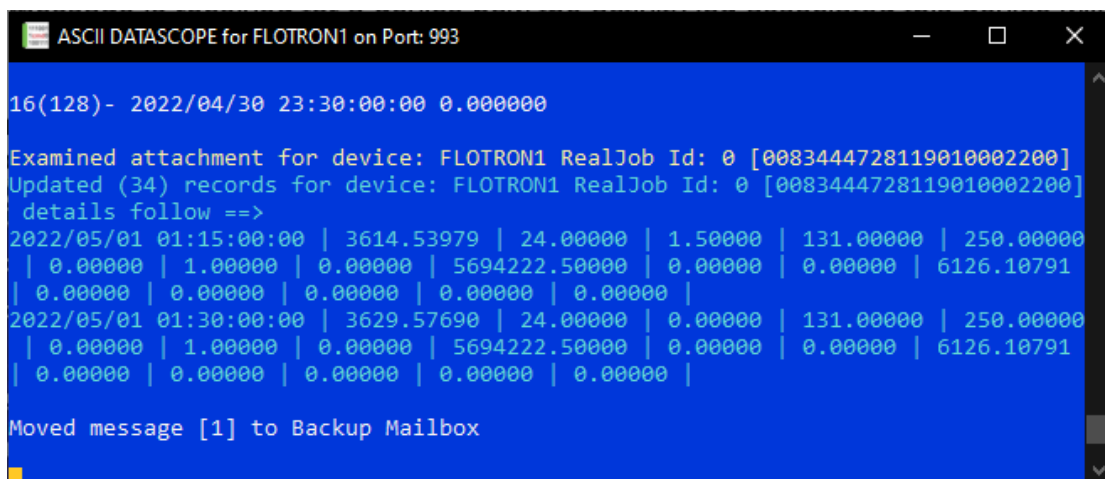
Let's say we have 2 devices, the first one will show activity of the mailbox, such as checking the inbox and displaying empty, etc. We will see updates for that device in white and blue as data. The next datascope opened at the same time will only display its own activity, which will become active when a valid mail is received, and the driver begins processing the data. Refer to the following diagrams for more information:





Now we will receive 3 messages to the inbox: an email for FLOTRON1 (currently started), an email for FLOTRON2 (currently stopped) and a spam message.

Note when a message is received for a device how the activity log associated with this device will display the data being collected from the email. In this case, an email has been identified for FLOTRON1, so the message unpacking is displayed on FLOTRON1's activity log. Once completed, this message is moved to the mailbox listed in the device configuration for the Backup Mailbox (in this case BACKUPBOX).



Now when the message for FLOTRON2 has been detected, it will not attempt to parse the message as the device is set to stopped in Adroit. This email is instead moved to the mailbox listed in the device configuration for the ToDo Mailbox (in this case TODOBOX).

```

ASCII DATASCOPE for FLOTRON1 on Port: 993

IMAP Mailbox Empty...
IMAP Mailbox Empty...

Date[Wed, 16 Nov 2022 10:25:21 +0200] From["Adroit Drivers" <adroitemaildriverte
st@gmail.com>] Subject[F00001 : 0083444728119010002201]

Receive of 1 Message(s) successfull, will now attempt to parse and validate and
update adroit appropriately.

Raw from _CSList --> Date[Wed, 16 Nov 2022 10:25:21 +0200]Subject[F00001 : 00834
44728119010002201]

[2022-05-01<SP>01:15:00,3614.53979,24,00000,1.50000,131.00000,250.00000,0.00000,
1.00000,0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000
,0.00000,0.00000]
[2022-05-01<SP>01:30:00,3629.57690,24,00000,0.00000,131.00000,250.00000,0.00000,
1.00000,0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000
,0.00000,0.00000]
[Checksum:<SP>29208]

Moved unconfigured message [1] to Todo Mailbox(content not validated)

```

The example spam message will be moved to the mailbox listed in the device configuration for the Error Mailbox field (in this case ERRORBOX).

```

ASCII DATASCOPE for FLOTRON1 on Port: 993

IMAP Mailbox Empty...
IMAP Mailbox Empty...

Date[Wed, 16 Nov 2022 10:26:22 +0200] From["Adroit Drivers" <adroitemaildriverte
st@gmail.com>] Subject[THIS IS AN EXAMPLE OF A SPAM MESSAGE]

Receive of 1 Message(s) successfull, will now attempt to parse and validate and
update adroit appropriately.

Moved errored message [1] to Error Mailbox(content not validated)

IMAP Mailbox Empty...

```

Now when FLOTRON2 is set to started in Adroit, the driver will allow any emails that are for that device to start being read into Adroit. Any emails in the ToDo Mailbox need to be manually moved from that mailbox back into the Inbox.

FLOTRON2 is now started, and the email has been moved manually back to the Inbox, the message will be read. Note that the data read in the email will appear on the datascope for FLOTRON2, but again indication of the activity is also noted on FLOTRON1.

```
Select ASCII DATASCOPE for FLOTRON1 on Port: 993

IMAP Mailbox Empty...

Date[Wed, 16 Nov 2022 10:25:21 +0200] From["Adroit Drivers" <adroitemaildriverte
st@gmail.com>] Subject[F00001 : 0083444728119010002201]

Receive of 1 Message(s) successfull, will now attempt to parse and validate and
update adroit appropriately.

Raw from _CSList --> Date[Wed, 16 Nov 2022 10:25:21 +0200]Subject[F00001 : 00834
44728119010002201]

[2022-05-01<SP>01:15:00,3614.53979,24.00000,1.50000,131.00000,250.00000,0.00000,
1.00000,0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000
,0.00000,0.00000]
[2022-05-01<SP>01:30:00,3629.57690,24.00000,0.00000,131.00000,250.00000,0.00000,
1.00000,0.00000,5694222.50000,0.00000,0.00000,6126.10791,0.00000,0.00000,0.00000
,0.00000,0.00000]
[Checksum:<SP>29208]

Moved message [1] to Backup Mailbox

IMAP Mailbox Empty...
```

```
ASCII DATASCOPE for FLOTRON2 on Port: 993

Examined attachment for device: FLOTRON2 RealJob Id: 0 [0083444728119010002201]
Updated (34) records for device: FLOTRON2 RealJob Id: 0 [0083444728119010002201]
details follow ==>
2022/05/01 01:15:00:00 | 3614.53979 | 24.00000 | 1.50000 | 131.00000 | 250.00000
| 0.00000 | 1.00000 | 0.00000 | 5694222.50000 | 0.00000 | 0.00000 | 6126.10791
| 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 |
2022/05/01 01:30:00:00 | 3629.57690 | 24.00000 | 0.00000 | 131.00000 | 250.00000
| 0.00000 | 1.00000 | 0.00000 | 5694222.50000 | 0.00000 | 0.00000 | 6126.10791
| 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 |
```

In summary:

- Any emails that fail to validate due to formatting errors, such as missing the checksum field at the end of an email, will be moved to the Error Mailbox.
- Any emails that are for an Adroit device that is stopped or does not exist will be moved to the ToDo Mailbox.
- Any emails that are for an Adroit device that is started, succeeds validation and has been parsed to Adroit will be moved to the Backup Mailbox.

5.2. Other Considerations

The not-healthy flag on a device will only be cleared once some valid data has been received for it, so on Agent server start-up it will show unhealthy until some valid email comes in, if after a while it still shows unhealthy then there is a problem.

5.3. Datalogging Considerations

Datalogging to Adroit proprietary database (LGD) files require special consideration.

Under normal circumstances, if the value does not change but the timestamp does change the value will not be logged if it is newer than the last time on that tag in question. However, historical data values will be logged as received even though there is no change. This must be done to allow the proper operation of historical logging.

If there is a requirement to log every value for every period you must force this by using Demand mode with "self" triggering, A1 triggers itself. This will force ALL values to be logged regardless of whether they have changed.

Edit Datalog agent: MYCITY_A1

Logging
 Slot: ☐ Header slots ☐ Start ☒ Stop
 value

Mode
☐ Continuous ☒ Demand ☐ Both

Trigger: MYCITY_A1.value

Log Now

OK
 Retrieve...
 Advanced...
 Help
 Header...

Data Set Configuration
 Set number: 0 Add Delete
 Datasource: C:\ProgramData\Adroit Technologies\Adroit\Configural
 Length... 10day (86400 samples, 1728000 bytes)

Deadbands
 Time... 10sec
 Value: 0.0000

Out of limit logging
 Time... 1sec
 Upper limit: 0.0000
 Lower limit: 0.0000

Compression
 Method: Actual

☐ Disable logging on standby

Options (global to all sets)
☐ Clear at start
☐ Interpolate
☐ Double entries
 Backup period: Disabled
 Delete backup files older than (days): 90
☐ Enable CSV backups

Clicking **Advanced**:

To stop logging of startup values that happen at the time of startup, enable the **"Disable first datalog entries on startup of agent server (SuppressLogAtStartup = 1)"** checkbox.

Advanced Datalog Configuration (global settings)

General:
 Datalog system buffer size (Kb): 500
☒ Disable legacy file compression of 5:1 (LogCompressionFactor = 1)
☒ Flush lazily (LazyFlushing = 1)
☒ Disable first datalog entries on startup of agent server (SuppressLogAtStartup = 1)
☐ Support compression algorithm on demand logging (SupportCompressionOnDemand = 1)

SQL logging:
☒ Check for existence of database tables on startup (CreateTableOnStartup = 1)
☐ Disable database table name prefixes (DisableADOTableNamePrefix = 1)
☐ Add normal timestamp fields to database tables (LogADODDataTimes = 1)
☐ Use only proprietary logged data for trends (ProprietarySetPreference = 1)

UI settings (no restart required):
☒ Create datalog agents automatically (AutoCreate = 1)

OK Cancel

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

We recommend if you have any queries or issues, please visit our Knowledge Base for more information and additional troubleshooting. If you still cannot find a resolution for your problem, please contact Adroit Support.

Adroit Knowledge Base: <https://adroittechnologiesautomation.com/>

6.3.1. Cannot install the driver (Adroit 10.0.5.2 and earlier)

As of Adroit 10.0.5.2 Update 1, the MyCity Email driver does not have a .inf which is used to add the installation of the driver. To install the driver, please go to the following Adroit directory and move the mycity.dll from the **Drivers** folder into the main directory. So, move the mycity.dll from:

C:\Program Files (x86)\Adroit Technologies\Adroit\Drivers

Move into the directory:

C:\Program Files (x86)\Adroit Technologies\Adroit

6.3.2. Access denied error

Error code according to the device datascope:

[10013] Access denied

[10013] Access deniedCannot do email work!

If you are experiencing this error, please check that the following are correct:

- Login Name and Password
- Mail Server and Mail Port

If this is still not causing the issue, please make sure that your firewall is not currently blocking the port that was provided in the device configuration.

6.3.3. I cannot connect with my Gmail account

As of May 2022, Google has enforced stricter security access for 3rd party applications that have access to a Google account and want to make use of their services (such as Gmail). A Knowledge Base article has been posted explaining how to configure an App Password to allow Adroit access to your Gmail account (<https://adroittechnologiesautomation.com/t/gmail-setup-for-email-drivers-using-imap-new-google-security-policy/1304>).

That same article is in [Appendix A: Gmail Setup for Email Drivers using IMAP \[New Google Security Policy\]](#) section of this document.

7. Appendix A

7.1. Gmail Setup for Email Drivers using IMAP [New Google Security Policy]

This knowledge base article details the steps required in configuring the Gmail IMAP server with the mentioned Adroit email drivers, in line with Gmail's new security policies implemented as of **30 May 2022**. For further information, refer to the following Google Account Help article: <https://support.google.com/accounts/answer/6010255?hl=en>

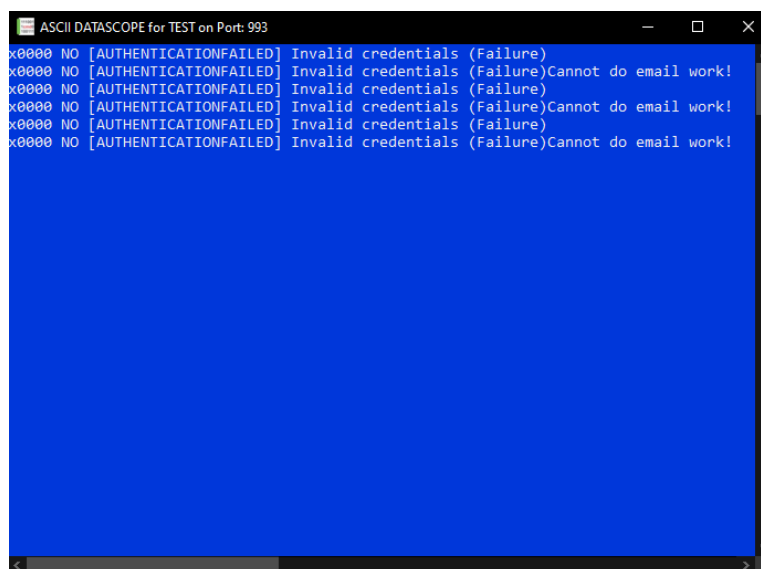
The opening article states the following:

To help keep your account secure, from **May 30, 2022**, Google no longer supports the use of third-party apps or devices which ask you to sign in to your Google Account using only your username and password.

Important: This deadline does not apply to Google Workspace or Google Cloud Identity customers. The enforcement date for these customers will be announced on the Workspace blog at a later date.

For more information, continue to read.

Previously, Adroit email driver's required full access to your Gmail account, and subsequently your Google account, to allow access into your inbox using the Gmail IMAP server. This required adjusting a security setting on your Google account to allow 3rd party applications to access your account. If you have noticed that Adroit has not been able to successfully connect to the Gmail IMAP server, typically stating an authentication error, the problem could be related to this issue.

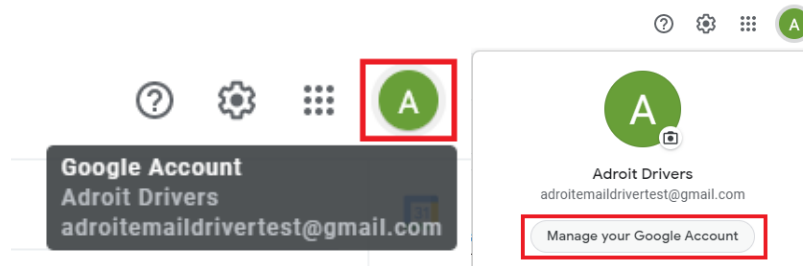


Example of MYCITY Email Client Protocol Driver device authentication error [x0000 NO [AUTHENTICATIONFAILED] Invalid Credentials (Failure) Cannot do email work!]

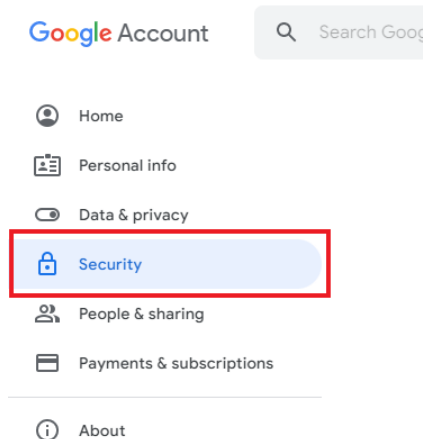
To access the setting in question, follow the steps provided:

1. Accessing from Gmail directly:
 - a. Log in to your Gmail account.

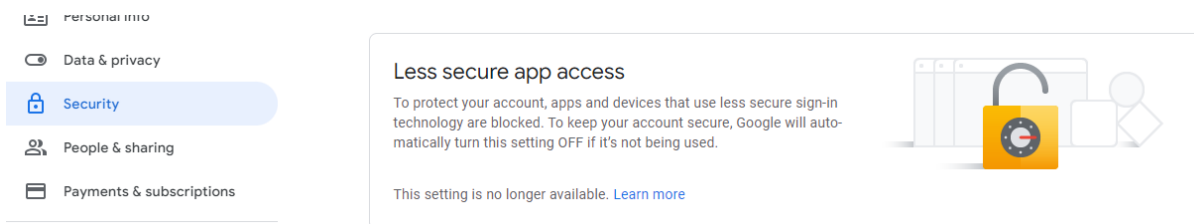
- b. Click on the Google Account indicator in the top right-hand corner of the screen. Click on **Manage your Google Account**



- c. On the page that opens, click on the Security tab located on the left-hand side.



- d. Under the Security section, there used to be a section Less secure app access which was required to be On. However, this setting has since been removed.



PLEASE NOTE

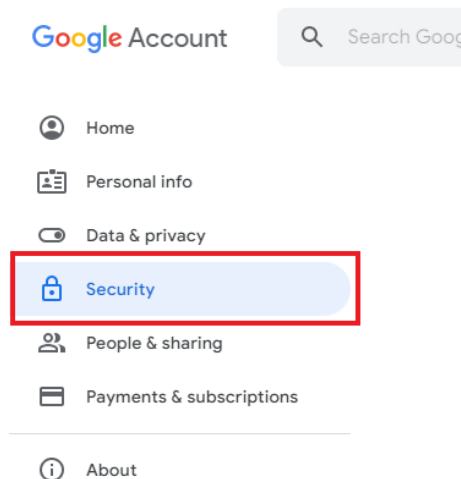
If you are unable to see this setting, you probably have 2-Step Verification turned on. In the past, Adroit would require this be turned off, but please read further as this is now required to be on.

2. Accessing account security directly

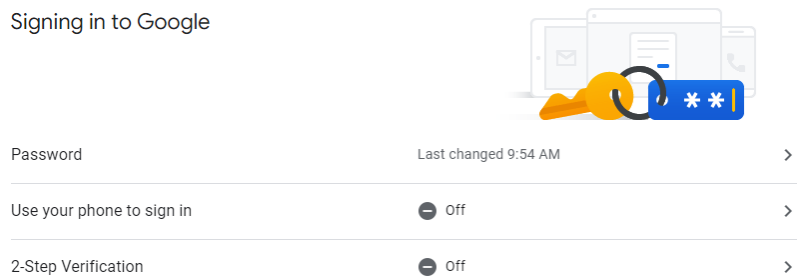
- Log in to your Google Account
- Access your Google Account: <https://myaccount.google.com/>
- Follow from **Step 1.c** outlined previously.

To fix this issue, the user will need to create an App Password that Adroit will then use for authentication with Gmail. This requires that 2-Step Verification is enabled. The steps provided below are summarised and can also be read from here, should you require additional information: <https://support.google.com/accounts/answer/185833?hl=en#:~:text=Create%20%26%20use%20App%20Passwords>

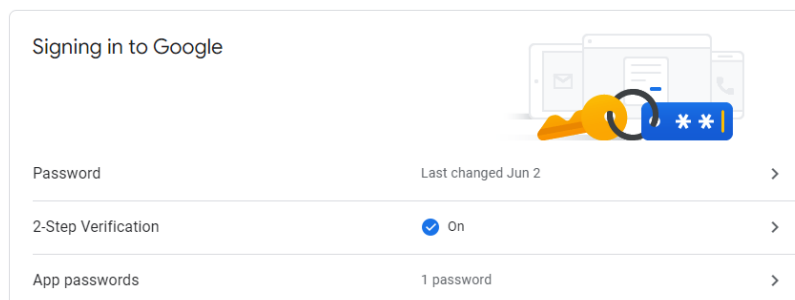
1. **Log in** to your Google/Gmail Account
2. Access your **Google Account**: <https://myaccount.google.com/>
3. Click on the **Security** tab on the left-hand side of the screen



4. Navigate the menu to the section **Signing in to Google**.



5. Enable **2-Step Verification** if it is not already. If has not been enabled, do so and follow the prompts by Google. You will need a cellphone number to continue.
6. After setting up 2-Step Verification, click on the new option under the **Signing in to Google** called **App passwords**.



7. After verifying it is you, you will see a list of App passwords currently registered. Generate a new App password by clicking the **Select app** dropdown box, and selecting the **Mail** option, then the **Select device** dropdown box, selecting the Windows Computer option. Click on the **Generate** button.

Select the app and device you want to generate the app password for.

Mail

Select device

iPhone

iPad

BlackBerry

Mac

Windows Phone

Windows Computer

Other (Custom name)

GENERATE

8. A message will appear with a 16-character randomly generated password. Copy this and save it somewhere secure. This is the password you will use for Adroit to access the Gmail account.

Generated app password

Your app password for Windows Computer

•llg a,fl wakj mojc

How to use it

1. Open the "Mail" app.
2. Open the "Settings" menu.
3. Select "Accounts" and then select your Google Account.
4. Replace your password with the 16-character password shown above.

Just like your normal password, this app password grants complete access to your Google Account. You won't need to remember it, so don't write it down or share it with anyone.

[Learn more](#)

Add your Google account

Enter the information below to connect to your Google account.

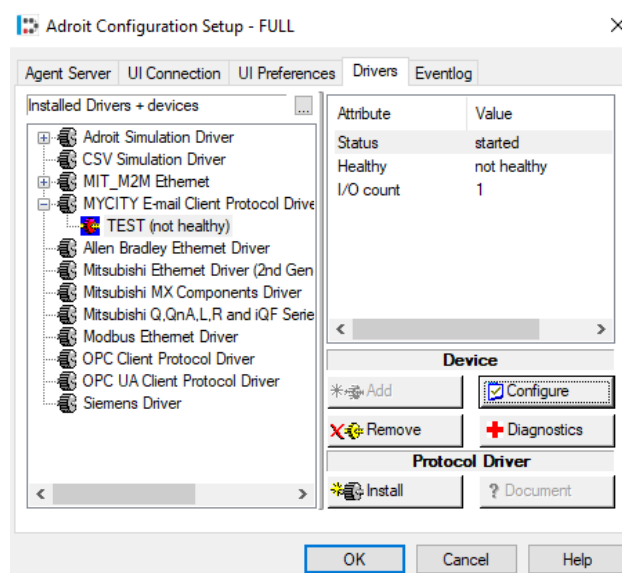
Email address
seuresally@gmail.com

Password
••••••••

☐ Include your Google contacts and calendars

DONE

9. Open the **Classic Adroit Setup** and navigate to the device experiencing the issue.
10. Open the device configuration by selecting the device and clicking **Configure**.



11. Ensure the details of the device have been input correctly and place the App password previously generated into the **Login Password/Password** input field.

MYCITY E-mail Client Protocol Driver : TEST

Port Selection
☐ Enable Secondary
☒ Primary
☐ Secondary

Local Device Settings
 Station Id: 0083444728119010002200

Common Mail Settings (*Resource key)
 Resource: 3
 *Mailbox Type: ☒ IMAP ☐ POP
 *Mail Server: imap.gmail.com
 *Mail Port: 993
 Login Name: adroitemaildriverstest2@gmail.com
 Login Password: [Masked]
 Mailbox Poll Period (ms): 2000
☒ Move Used/ToDo Messages to Backup/ToDo Mailboxes.
☐ Delete Used/ToDo Messages.
☐ Delete Spam Messages.
☒ Server uses SSL.
☒ Detailed Debugging on Datascope.
 Incoming Mailbox: INBOX
 Backup Mailbox: BACKUPBOX
 ToDo Mailbox: TODOBOX
 Error Mailbox: ERRORBOX

OK Cancel

12. Exit the Device configuration by clicking on OK. You may be prompted to change the current resource for all mailboxes if you are using the MyCity Email driver, for example. You can create a new resource or override it for all devices using the same resource.

13. The device should now be marked as healthy. Open the datascope to see if the error is persisting.



PLEASE NOTE

You may need to restart your Agent Server before the changes are applied.



PLEASE NOTE

The App password generated to allow access to Gmail's Email Server does not rely on the verification that 2-Step Verification provides, as it allows the third-party application limited control direct access to your Gmail account. Please keep this password private and secure.

However, considering these changes, Google may continue to further change and evolve their security policies, with or without notice, which can affect the ability of the Adroit email drivers to function as expected. It is therefore recommended that users provide an alternative email with a different provider as a backup solution, or to migrate to another email provider.

Provided is a list of tested email providers that can be utilized with Adroit email drivers as an alternative to Gmail, and more will be added in the future:

- Outlook

Please contact Adroit Support should you wish to enquire about compatibility with other email service providers.

For further details on device configuration, please refer to the respective Adroit driver documentation.

For any additional information or queries, please feel free to contact [Adroit Support](#) or search our [Knowledge Base](#).

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
13	06-Jun 2022	Added ignore Checksum checkbox to configuration dialog.
14	28-Jun 2022	Bug fixes.